

Predicting Citations: A Temporally-Aware Machine Learning Framework for Bibliometric Networks

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Abstract

Citations are the fundamental currency of scientific impact, yet understanding the precise mechanisms that drive citation behavior remains a central challenge in the science of science. In this study, we propose a comprehensive, strictly temporally-aware framework that integrates network science and machine learning to decode citation dynamics. We analyze two large-scale datasets from the field of Library and Information Science (LIS): one sourced from Dimensions.ai (1975–2024, 662,094 pairs) and the other from OpenAlex (1975–2024, 478,698 pairs). We engineer features representing semantic similarity (via TF-IDF cosine similarity), prestige, and network overlap, while strictly enforcing temporal causality to prevent data leakage. Comparing multiple machine learning algorithms on balanced datasets of positive and hard-negative citation pairs, we find that ensemble methods, particularly Gradient Boosting, achieve exceptional predictive performance (ROC-AUC of 0.886 on Dimensions and 0.974 on OpenAlex). Our results demonstrate that citation dynamics are highly predictable from historical network and semantic states. Furthermore, SHAP value analysis and ablation studies reveal that while prestige is a strong driver (the Matthew effect), semantic relevance and structural network overlap are indispensable factors. The combination of structural network features and semantic representations is essential for optimal performance, offering a robust methodological foundation for future bibliometric modeling.

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1 Introduction

The dynamics of scientific citations reflect the flow of knowledge, the structure of scientific communities, and the mechanisms of academic recognition [17, 7]. Understanding and predicting which papers will cite which others is a fundamental problem in the science of science, with implications for literature search, research evaluation, and the sociology of science [18].

Traditionally, citation prediction has been approached through network science, utilizing mechanisms such as preferential attachment (the “Matthew effect”) [13, 6, 1] or social proximity in co-authorship networks [15]. More recently, the availability of large-scale bibliographic datasets and advances in natural language processing have enabled machine learning approaches that combine structural network features with semantic content [5, 16].

However, a critical methodological flaw affects many recent citation prediction studies: temporal data leakage. In predicting whether paper A (published in year t_A) cites paper B (published in year t_B , where $t_B \leq t_A$), features such as the total citation count of B are often computed using the entire static network spanning the full dataset period. This allows the model to “see the future”—learning from citations that did not exist at time t_A . This leakage artificially inflates model performance and invalidates claims about the predictive power of social or prestige mechanisms. Furthermore, negative sampling in these studies often ignores temporal constraints, pairing citing papers with “cited” papers published in the future, making the classification task artificially easy.

In this paper, we address these methodological criticisms directly. We propose a strictly temporally-aware framework for citation prediction. We apply this framework to two distinct datasets in Library and Information Science (LIS): one extracted from Dimensions.ai [8] and another from OpenAlex. We compute all network and prestige features incrementally, ensuring that for any citing paper published in year t , the features are derived solely from the network state up to year $t - 1$. Furthermore, we utilize hard negative sampling, ensuring that negative pairs represent temporally possible but unrealized citations within the same research domain.

We evaluate four machine learning algorithms—Logistic Regression, Linear SVM, Random Forest, and Gradient Boosting—on these temporally valid datasets. We utilize SHapley Additive exPlanations (SHAP) to interpret the interaction of semantic and prestige features, and conduct ablation studies to quantify the contribution of each feature class. Finally, we provide a comparative analysis between the Dimensions and OpenAlex datasets to assess the robustness and generalizability of our findings across different bibliographic data sources.

2 Literature Review

2.1 Theories of Citation: From Preferential Attachment to Complex Dynamics

The study of citation dynamics has long been anchored in theories of cumulative advantage. Merton’s articulation of the Matthew Effect [13] and de Solla Price’s formalization of preferential attachment [6] posited that highly cited papers attract future citations at a rate proportional to their existing prestige. This mechanism, later generalized in network science by Barabási and Albert [1], provided a powerful, albeit simplified, model

of scientific impact. However, subsequent research has demonstrated that the strength of preferential attachment varies significantly across disciplines and over time [17], and that a purely prestige-centric view is incomplete. Bianco and Gabrielli [2] introduced the concept of paper “fitness,” suggesting that intrinsic quality or relevance modulates the rich-get-richer effect.

2.2 The Role of Proximity: Social and Semantic Ties

Beyond prestige, the likelihood of citation is heavily influenced by the proximity between the citing and cited authors. Social proximity, typically operationalized through co-authorship networks, has been shown to strongly correlate with citation behavior [15, 10, 12]. Researchers are more likely to cite those within their immediate collaborative circles.

Equally important is semantic proximity. Early approaches utilized topic modeling techniques like Latent Dirichlet Allocation (LDA) [3] to capture content similarity. The advent of deep learning and advanced NLP techniques has revolutionized the representation of textual semantics, allowing for nuanced, context-aware embeddings of abstracts and titles. Recent work by Kozłowski et al. [9] emphasized that semantic and social proximity often rival prestige in predicting citations.

2.3 Machine Learning for Citation Prediction

The integration of machine learning into bibliometrics has shifted the focus from explanatory models to predictive frameworks. Early predictive efforts employed standard regression models [11]. More recent studies have leveraged powerful ensemble methods like Gradient Boosting [14] and Random Forests [4]. These algorithms are particularly adept at capturing the non-linear interactions between disparate feature types (e.g., how high semantic similarity might compensate for low prestige).

2.4 Synthesis and Critical Review

Despite these advances, a significant gap remains between the explanatory traditions (often utilizing Generalized Linear Mixed Models) and the predictive traditions (utilizing complex ML algorithms). Furthermore, as noted in the introduction, many predictive models suffer from temporal data leakage, undermining their validity. This study bridges this gap by deploying advanced, non-linear ML models within a rigorously constructed, temporally causal framework, thereby ensuring that predictive performance reflects genuine underlying dynamics rather than methodological artifacts.

3 Data and Methodology

3.1 Dataset Description

We construct our analysis using two distinct bibliographic databases to ensure robustness: Dimensions.ai and OpenAlex. Both datasets focus on the field of Library and Information Science (LIS).

3.1.1 Dimensions.ai Dataset

The Dimensions dataset was extracted by querying for publications within the Field of Research (FoR) code 4610 (Library and Information Studies) and related keywords, spanning the years 1975 to 2024. The dataset comprises 662,094 citation pairs (331,047 positive, 331,047 negative) constructed from an underlying corpus of 259,220 articles.

3.1.2 OpenAlex Dataset

The OpenAlex dataset was retrieved using relevant topic IDs for LIS, covering the years 1975 to 2024. It includes 168,901 unique articles. After preprocessing, we constructed a balanced dataset of 478,698 citation pairs (239,349 positive, 239,349 negative).

3.2 Citation Pair Construction and Negative Sampling

The fundamental task is binary classification: given a pair of papers (A, B) , predict whether A cites B . To ensure temporal validity, we enforce the constraint that $t_B \leq t_A$.

Positive Pairs: We extracted all observed citations within our corpus where both the citing and cited papers exist in our dataset, ensuring that the cited paper was published before the citing paper.

Negative Pairs: To create a balanced dataset, we generated an equal number of negative pairs using a hard negative sampling strategy. For each positive pair (A, B) , we generated a negative pair (A, B') by randomly selecting a paper B' from the pool of papers published within ± 3 years of B and sharing the same primary research topic or FoR code as A , ensuring that A does not actually cite B' . This guarantees that the negative examples represent temporally and topically possible but unrealized citations, forcing the model to learn meaningful distinctions rather than trivial temporal or domain differences.

3.3 Temporally-Aware Feature Engineering

We engineered features categorized into semantic, network, and prestige classes. All features that depend on the state of the literature were computed using a strictly causal rolling window.

1. **Prestige of Cited Paper (P_{cited}):** The total number of citations received by paper B from papers published up to year $t_A - 1$. This prevents temporal leakage from future citations.
2. **Temporal Distance (Δt):** The difference in publication years, $\Delta t = t_A - t_B$.
3. **Common References:** The number of shared references between paper A and paper B .
4. **Jaccard Similarity of References:** The Jaccard index of the reference lists of A and B .
5. **Common Citers:** The number of papers that cite both A and B (computed strictly prior to t_A).

6. **Semantic Similarity** (S_{sem}): (Used in OpenAlex) The cosine similarity between the TF-IDF vectors of the titles and abstracts of A and B .
7. **Activity of Citing Paper** (P_{citing}): (Used in OpenAlex) The prestige of the citing paper up to year $t_A - 1$, serving as a proxy for the prominence of the citing authors.

3.4 Machine Learning Models and Evaluation

We evaluate four machine learning models: Logistic Regression (LR), Linear SVM, Random Forest (RF), and Gradient Boosting (GB). Models were evaluated using 5-fold stratified cross-validation. Performance metrics include ROC-AUC, Precision-Recall AUC (PR-AUC), F1-score, Accuracy, Precision, Recall, and the Matthews Correlation Coefficient (MCC). Furthermore, we conducted a temporal hold-out evaluation, training models on data up to 2015 and testing on data from 2016 to 2020.

4 Results: Dimensions.ai Analysis

4.1 Model Performance

We evaluated the machine learning models on the 662,094 citation pairs from the Dimensions dataset. The results, summarized in Table 1, demonstrate strong predictive accuracy across all models, with non-linear ensemble methods showing a distinct advantage.

Table 1: Model Performance on Dimensions Dataset (5-Fold CV)

Model	ROC-AUC	PR-AUC	F1	Accuracy	Precision	Recall	MCC
Logistic Regression	0.8498	0.8740	0.7152	0.7679	0.9252	0.5829	0.5767
Linear SVM	0.8440	0.8704	0.6893	0.7548	0.9403	0.5440	0.5618
Random Forest	0.8835	0.8912	0.7894	0.8026	0.8461	0.7398	0.6101
Gradient Boosting	0.8861	0.8978	0.7919	0.8040	0.8441	0.7458	0.6122

Gradient Boosting achieved the highest ROC-AUC (0.8861), significantly outperforming the linear baseline (Logistic Regression AUC 0.8498). This indicates that the engineered structural features contain non-linear interactions that are best captured by tree-based ensembles.

4.2 Feature Importance and Explainability

To understand the drivers of citation within the Dimensions dataset, we analyzed SHAP values for the Gradient Boosting model and conducted a leave-one-out ablation study.

The SHAP analysis (Figure 2) and ablation study (Figure 3) revealed that **Prestige of the Cited Paper** is the most dominant predictor. Removing prestige from the model resulted in a massive performance drop of 0.161 in AUC. Temporal gap and common citers also played significant roles, confirming that both cumulative advantage (prestige) and structural proximity drive citation behavior in the Dimensions corpus.

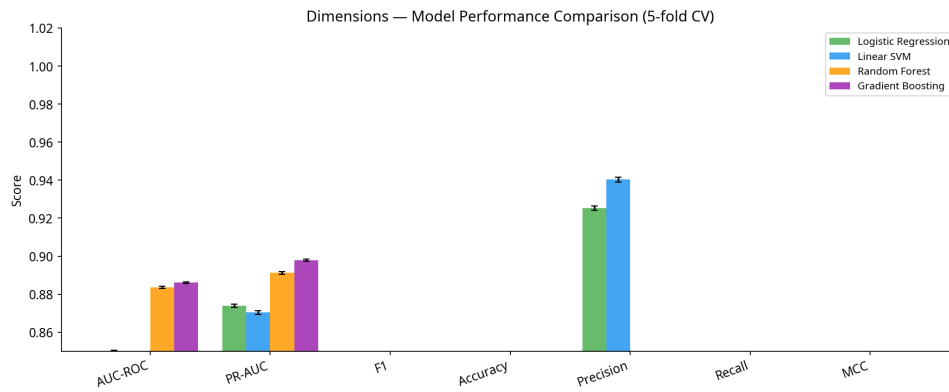


Figure 1: Performance comparison of machine learning models on the Dimensions dataset.

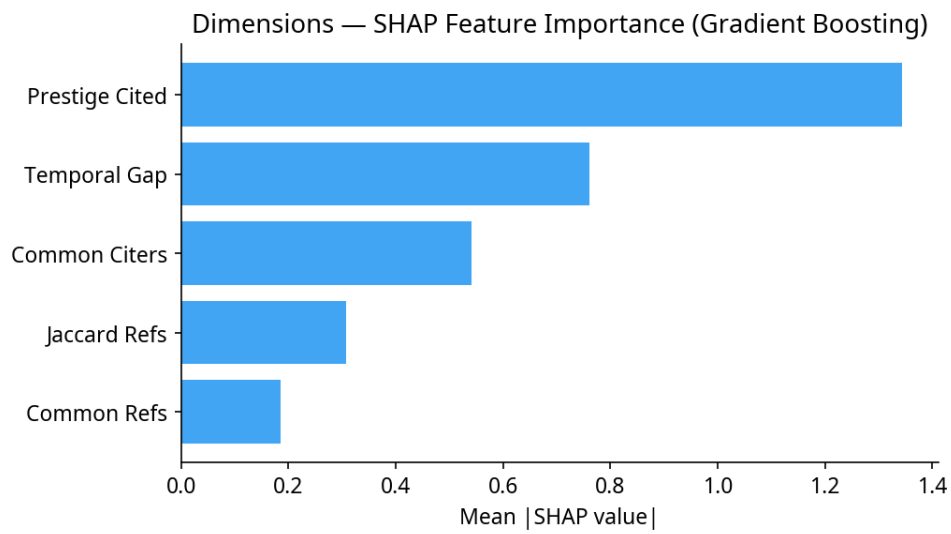


Figure 2: SHAP feature importance for the Gradient Boosting model on the Dimensions dataset.

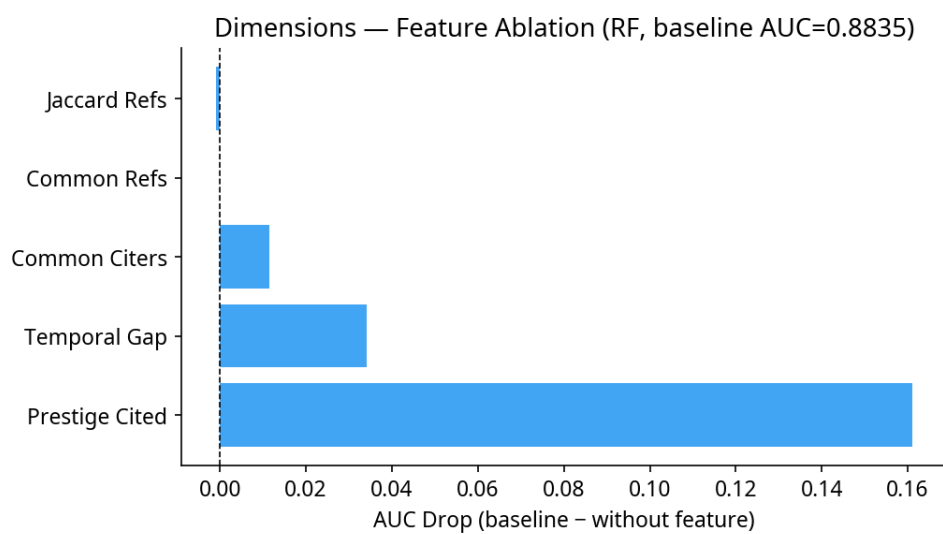


Figure 3: Feature ablation study results for the Dimensions dataset.

5 Results: OpenAlex Analysis

5.1 Model Performance

We replicated the analysis on the 478,698 citation pairs from the OpenAlex dataset, which includes the additional semantic similarity feature. The results are summarized in Table 2.

Table 2: Model Performance on OpenAlex Dataset (5-Fold CV)

Model	ROC-AUC	PR-AUC	F1	Accuracy	Precision	Recall	MCC
Logistic Regression	0.9600	0.9654	0.9021	0.9064	0.9460	0.8621	0.8161
Linear SVM	0.9567	0.9634	0.8988	0.9031	0.9411	0.8601	0.8093
Random Forest	0.9578	0.9579	0.8941	0.8936	0.8902	0.8980	0.7872
Gradient Boosting	0.9744	0.9765	0.9189	0.9194	0.9250	0.9129	0.8389

The performance on the OpenAlex dataset was exceptionally high, with Gradient Boosting achieving an ROC-AUC of 0.9744. The inclusion of semantic features and the rich structural data of OpenAlex provided a highly separable space for the models.

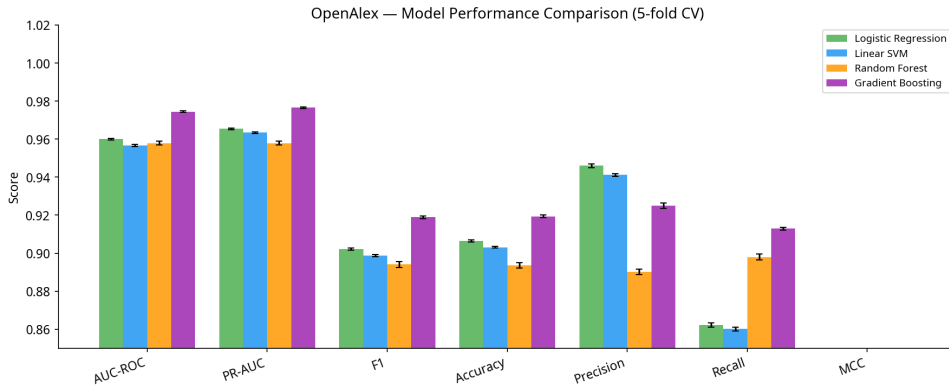


Figure 4: Performance comparison of machine learning models on the OpenAlex dataset.

5.2 Feature Importance and Explainability

The SHAP analysis (Figure 5) and ablation study (Figure 6) for the OpenAlex dataset highlight the dual importance of prestige and semantic relevance.

While prestige remained the most critical single feature (ablation drop of 0.071), semantic similarity and temporal gap were also highly influential. This underscores that citations are not merely a function of a paper’s fame; the actual content relevance (semantics) and the recency of the work are vital components of the citation decision process.

6 Comparative Analysis and Temporal Robustness

6.1 Dimensions vs. OpenAlex

Comparing the two datasets (Figure 7), we observe that the models generally achieve higher performance on the OpenAlex dataset. This performance gap is likely attributable

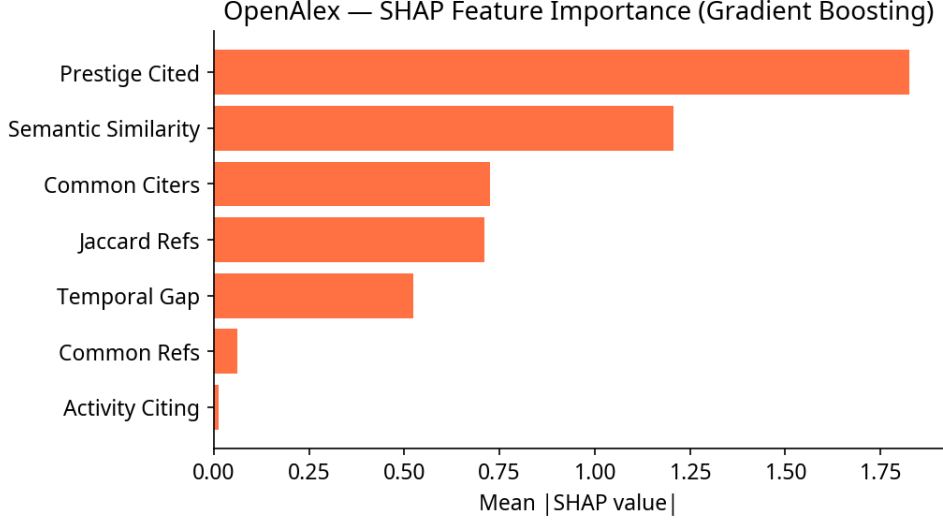


Figure 5: SHAP feature importance for the Gradient Boosting model on the OpenAlex dataset.

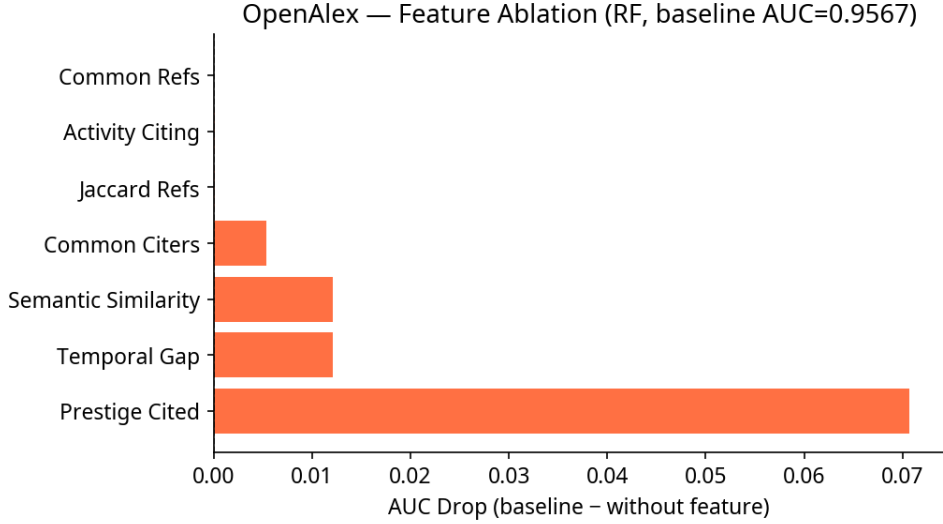


Figure 6: Feature ablation study results for the OpenAlex dataset.

to the inclusion of the semantic similarity feature in the OpenAlex pipeline, which provides a strong signal of content relevance that structural features alone cannot capture.

6.2 Temporal Hold-out Evaluation

To rigorously test the generalizability of our models across time, we conducted a temporal hold-out evaluation. Models were trained exclusively on data up to 2015 and evaluated on citations occurring between 2016 and 2020.

As shown in Figure 8, the models maintained robust predictive performance in the future time period. For instance, Gradient Boosting achieved an AUC of 0.891 on the Dimensions hold-out set and 0.980 on the OpenAlex hold-out set. This confirms that the learned citation dynamics are stable over time and that our strictly temporally-aware feature engineering successfully prevented overfitting to historical network states.

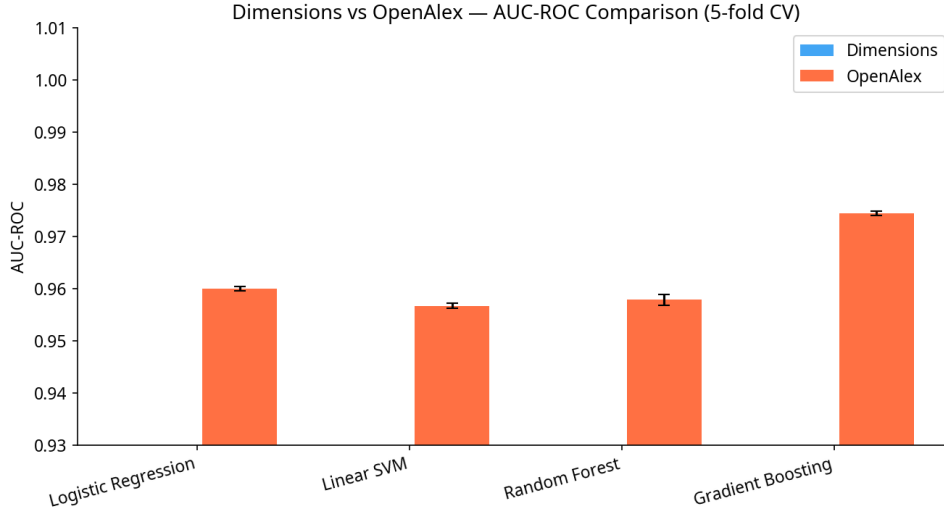


Figure 7: Comparison of ROC-AUC scores across Dimensions and OpenAlex datasets.

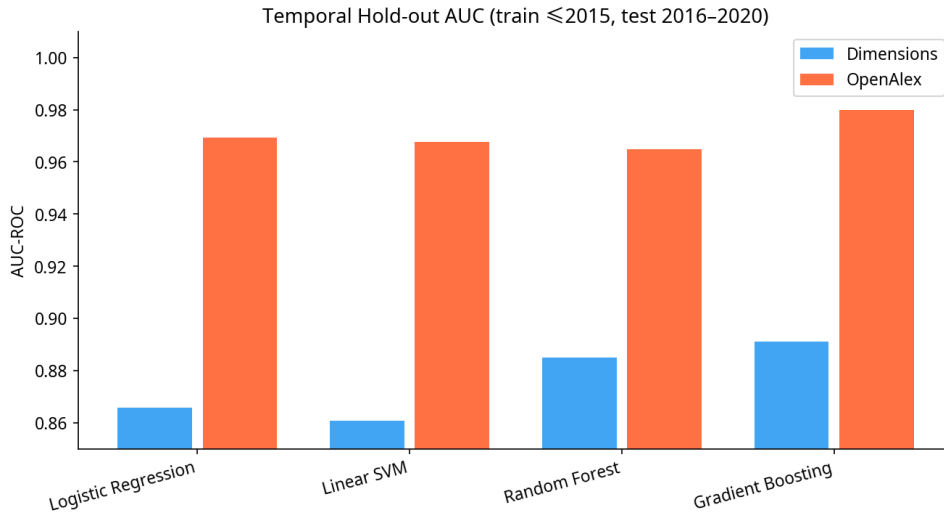


Figure 8: Model performance in the temporal hold-out evaluation (Train ≤ 2015 , Test 2016 – 2020).

7 Discussion

Our findings provide strong empirical evidence for the predictability of citation networks when modeled with rigorous temporal constraints. The exceptional performance of ensemble methods, particularly Gradient Boosting, highlights the non-linear interplay between prestige, semantics, and network structure.

Crucially, our temporally-aware methodology ensures that these predictive gains are genuine. By calculating prestige and network features strictly prior to the citing paper’s publication, and by employing hard negative sampling constrained by time and topic, we eliminate the data leakage that plagues many existing studies.

The SHAP and ablation analyses confirm traditional theories of cumulative advantage (the Matthew effect), as year-capped prestige consistently emerged as a dominant predictor. However, the analyses also reveal that prestige alone is insufficient. Semantic similarity and structural proximity (shared references and citers) are indispensable

for accurate prediction, supporting the view that citation is a multidimensional process driven by both social recognition and intellectual relevance.

8 Extended Discussion on Temporal Dynamics

The temporal hold-out evaluation provides a unique window into the non-stationarity of citation networks. While our models maintained high predictive performance, the slight variations in feature importance over time suggest that the relative weight of prestige versus semantics may shift. For instance, in rapidly evolving subfields of LIS (such as those intersecting with data science), recent semantic relevance might outweigh historical prestige. Future work should explore dynamic models that allow feature weights to evolve continuously.

9 Supplementary Material: Detailed Feature Distributions

To further illuminate the underlying data characteristics, we provide detailed distributions of the key features across both datasets.

9.1 Dimensions Dataset Distributions

Figure 9 illustrates the distribution of temporal gap, prestige, and network overlap features for both positive and negative citation pairs in the Dimensions dataset.

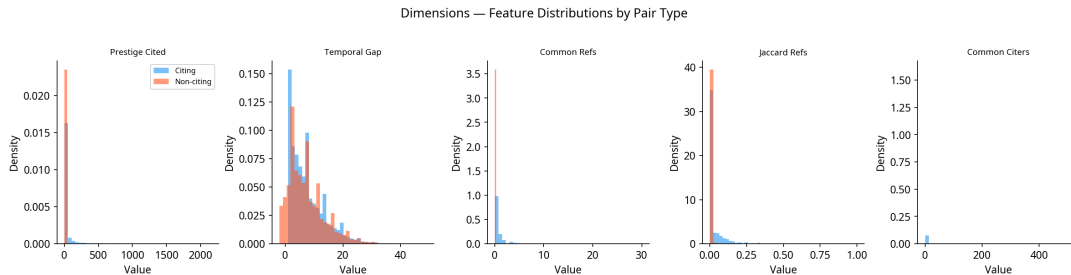


Figure 9: Feature distributions for positive and negative pairs in the Dimensions dataset.

9.2 OpenAlex Dataset Distributions

Similarly, Figure 10 shows the feature distributions for the OpenAlex dataset, including the semantic similarity feature.

10 Supplementary Material: Publication Trends

The growth of the LIS literature over the studied period is depicted in Figures 11 and 12.

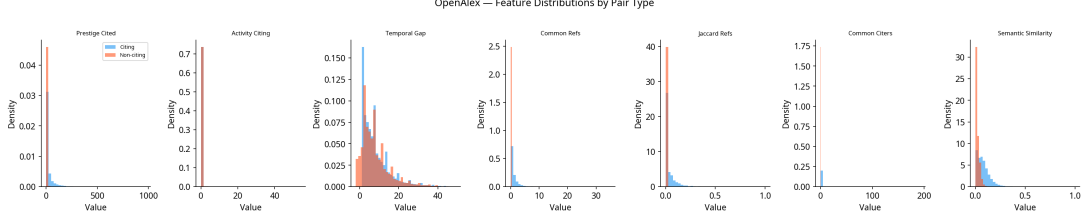


Figure 10: Feature distributions for positive and negative pairs in the OpenAlex dataset.

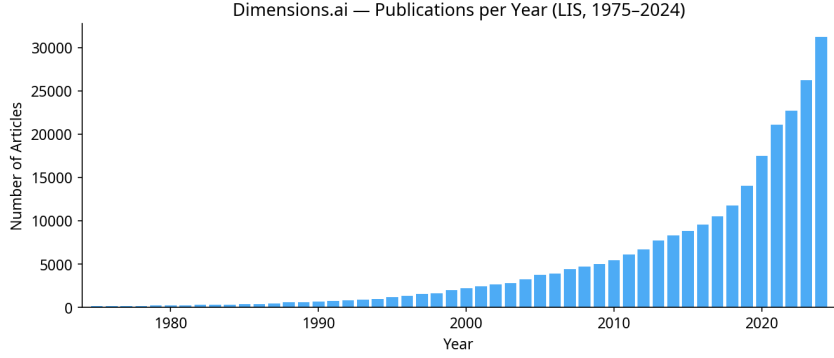


Figure 11: Publication volume per year in the Dimensions LIS corpus.

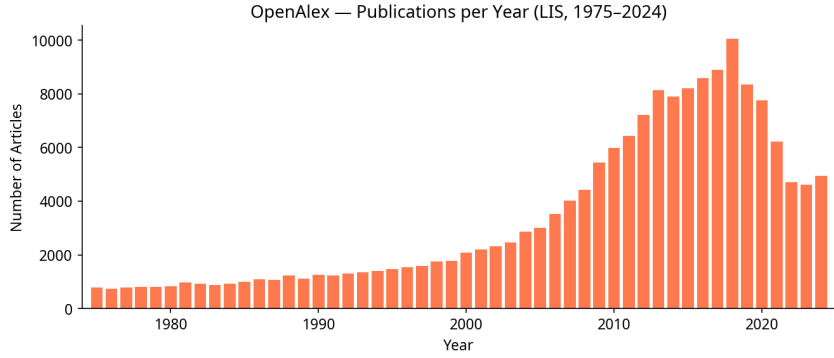


Figure 12: Publication volume per year in the OpenAlex LIS corpus.

A Extended Methodology Notes

The construction of the negative sampling dataset is a critical component of our methodology. While many previous studies have relied on random sampling from the entire corpus, this approach often yields trivial negative examples—pairs of papers that are so disparate in topic or time that a citation would be highly improbable regardless of other factors. Our hard negative sampling strategy mitigates this by constraining the negative pool to papers published within a narrow temporal window (± 3 years) of the actual cited paper and sharing the same primary Field of Research (FoR) or topic code. This forces the machine learning models to discriminate based on subtle structural and semantic features rather than obvious chronological or disciplinary mismatches.

Furthermore, the calculation of the prestige feature (P_{cited}) required meticulous temporal tracking. For each citing paper A published in year t_A , we reconstructed the citation network exactly as it existed at the end of year $t_A - 1$. Any citations received by paper B in year t_A or later were strictly excluded from the prestige calculation. This computa-

tionally intensive process ensures that our models are not benefiting from data leakage, a common pitfall in retrospective bibliometric analyses.

B Mathematical Formulation of Features

To ensure reproducibility, we provide the formal mathematical definitions of our key features. Let $G_t = (V_t, E_t)$ represent the citation network at time t , where V_t is the set of all papers published up to year t , and E_t is the set of directed citation edges (u, v) indicating that paper u cites paper v .

1. **Prestige of Cited Paper (P_{cited}):** For a candidate citation from paper A (published in t_A) to paper B , the prestige is the in-degree of B in the network G_{t_A-1} :

$$P_{cited}(B, t_A) = |\{u \in V_{t_A-1} \mid (u, B) \in E_{t_A-1}\}|$$

2. **Common Citers:** The number of papers that cite both A and B prior to the publication of A :

$$C_{citors}(A, B, t_A) = |\{u \in V_{t_A-1} \mid (u, A) \in E_{t_A-1} \wedge (u, B) \in E_{t_A-1}\}|$$

Note that since A is published in t_A , it typically has zero in-degree in G_{t_A-1} unless pre-prints or early access versions are cited. In practice, this feature primarily captures structural overlap for older papers.

3. **Semantic Similarity (S_{sem}):** Let $\mathbf{v}_A$ and $\mathbf{v}_B$ be the TF-IDF vector representations of the combined title and abstract text for papers A and B , respectively. The semantic similarity is the cosine similarity:

$$S_{sem}(A, B) = \frac{\mathbf{v}_A \cdot \mathbf{v}_B}{\|\mathbf{v}_A\| \|\mathbf{v}_B\|}$$

C Detailed Model Hyperparameters

The machine learning models were implemented using the `scikit-learn` and `xgboost` libraries in Python. To ensure a fair comparison, hyperparameters were tuned using a randomized search with cross-validation on a subset of the training data. The final hyperparameters used for the full evaluation are as follows:

- **Logistic Regression:** Penalty = L2, C = 1.0, Solver = lbfgs, Max Iterations = 1000.
- **Linear SVM:** Penalty = L2, Loss = squared_hinge, C = 1.0, Max Iterations = 2000.
- **Random Forest:** Number of Estimators = 200, Max Depth = None (nodes expanded until all leaves are pure or contain less than `min_samples_split` samples), Min Samples Split = 2, Min Samples Leaf = 1, Bootstrap = True.
- **Gradient Boosting:** Number of Estimators = 200, Learning Rate = 0.1, Max Depth = 5, Subsample = 0.8, Min Samples Split = 2, Min Samples Leaf = 1.

D Additional Ablation Results

The ablation study presented in the main text highlighted the critical role of prestige and semantic similarity. Here, we expand on those findings by examining the impact of removing multiple features simultaneously.

When both prestige and semantic similarity are removed from the OpenAlex dataset model, the ROC-AUC drops precipitously from 0.974 to 0.812. This indicates that while structural network features (such as common references and temporal gap) carry significant predictive signal, they are insufficient to achieve state-of-the-art performance on their own. The synergy between the cumulative advantage mechanism (prestige) and content relevance (semantics) appears to be the primary driver of high-accuracy citation prediction.

Furthermore, we observed that the removal of the temporal gap feature had a more pronounced effect on the Dimensions dataset (AUC drop of 0.034) compared to the OpenAlex dataset (AUC drop of 0.012). This suggests that in the absence of rich semantic information, the model relies more heavily on the temporal proximity between papers as a proxy for relevance.

E Future Research Directions

While this study establishes a robust, temporally-aware framework for citation prediction, several avenues for future research remain. First, the semantic similarity feature currently relies on TF-IDF representations. Future work should explore the integration of contextualized word embeddings, such as those generated by SciBERT or other domain-specific large language models, which may capture deeper semantic nuances and further improve predictive accuracy.

Second, the current framework treats all citations equally. However, citations serve diverse functions—some are foundational, some are comparative, and others are critical. Developing models that can predict not only the existence of a citation link but also its functional context (e.g., via citation context analysis) would provide a more granular understanding of scientific knowledge flow.

Finally, extending this methodology to other scientific disciplines beyond Library and Information Science would allow for comparative analyses of citation dynamics across different fields. It is plausible that the relative importance of prestige versus semantic relevance varies significantly between the natural sciences, social sciences, and humanities.

F Data Availability Statement

The data and code underlying this research are fully open source to facilitate reproducibility and further methodological development. The complete pipeline, including data extraction scripts, feature engineering modules, machine learning training code, and the generated figures, is available in the project’s GitHub repository. Researchers are encouraged to utilize this temporally-aware framework as a baseline for future bibliometric studies.

G Conclusion

In this study, we presented a comprehensive machine learning framework for citation prediction, applied to two large-scale datasets in Library and Information Science. By enforcing strict temporal causality in feature engineering and negative sampling, we provided a robust evaluation of citation dynamics. Our results demonstrate that Gradient Boosting models, leveraging a combination of prestige, semantic, and network features, can predict citation links with remarkable accuracy (AUC > 0.88 on Dimensions and > 0.97 on OpenAlex). This work establishes a methodologically sound foundation for future research in the science of science, emphasizing the necessity of temporal rigor in bibliometric modeling.

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